

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Observation Task No.: 2

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Date: 28/7/26

1. TITLE

Working with CSS3 and HTML5 Canvas Graphics

2. AIM

To understand how CSS3 enhances the appearance of web pages and explore HTML5 graphics using the Canvas element.

3. OBJECTIVE

The objectives of this observation task are:

- To study page visual presentation enhancement using CSS3 styling structures.
 - To apply border styles, rounded corners (border-radius), and drop shadows (box-shadow).
 - To implement Flexbox container models for layout positioning and grid wraps.
 - To utilize the HTML5 <canvas> tag to render inline 2D raster vector drawings.
 - To write scripting logic executing paths, shapes fills, outlines, and text strokes on canvas.
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4. THEORY

CSS3 (Cascading Style Sheets, Level 3) extends the visual presentation layer of HTML pages, offering rules to align, format, and animate elements. Flexbox organizes card blocks in container alignments, while drop shadows and border corners add layers of depth. HTML5 introduces graphics natively via the <canvas> element, which exposes a 2D rendering context. Javascript scripts target this canvas API to programmatically draw rectangles, lines, arcs, and text on a grid coordinate layout.

Theory of CSS3 Layouts & HTML5 Canvas:

- **CSS3 Selectors & Transitions:** Box elements are styled using *background-color*, *border-radius*, and *box-shadow*. Interactive hover transitions are applied using the *transition* shorthand.
 - **CSS3 Flexbox:** Enforces standard container alignments (*display: flex*) dynamically adjusting spacings (*gap*) and wrapping grid elements.
 - **HTML5 Canvas Element:** A raster graphics container. The context object (*canvas.getContext('2d')*) exposes drawing commands.
 - **2D Context Drawing Operations:** Rectangles are drawn using *fillRect()* and *strokeRect()*. Paths are initiated with *beginPath()*, mapped with *moveTo()* and *lineTo()*, and drawn with *stroke()*. Arcs are drawn via *arc()*.
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5. ALGORITHM / PROCEDURE

1. Establish project directories: *Observation/ObTask2/html* and *Docs*.
 2. Create *canvas.html* to implement graphic scripts.
 3. Embed a `<canvas>` tag with custom dimensions and borders.
 4. Retrieve the 2D canvas context and execute *fillRect*, *strokeRect*, and line path strokes.
 5. Create *partB.html* to implement CSS3 enhancements.
 6. Construct layout containing headers, nav sections, and a flex container hosting three card blocks.
 7. Apply CSS rules: flex displays, card dimensions, margins, rounded corners, box shadows, and hover transitions.
 8. Open files in browser, verifying CSS hover properties, Flexbox wrap alignments, and Canvas graphics rendering.
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6. PROGRAM

Files: *canvas.html* and *partB.html* (Part 1)

canvas.html (Complete) & partB.html - Part 1 (Lines 1-50)

```
// --- canvas.html ---
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Styled WebPage</title>
</head>
<body>
    <center><canvas height="500px" width="1000px" id="canvas" style="border: 1px solid black; border-radius:
5px;"></canvas></center>
</body>
<script>
const canvas = document.getElementById("canvas")
const ctx = canvas.getContext("2d")
ctx.fillStyle = "blue"
ctx.fillRect(20,20,150,200)
ctx.strokeStyle = "red"
ctx.lineWidth = 5
ctx.strokeRect(200,20,150,100)
ctx.beginPath();
ctx.moveTo(50,300)
ctx.lineTo(200,300)
ctx.strokeStyle = "green"
ctx.lineWidth = 5
ctx.stroke()
ctx.beginPath()
ctx.arc(300,200,50,0,Math.PI*2)
ctx.fill()
ctx.beginPath()
ctx.fillText("KING CYCLONE BY SRIKAR", 500, 250);
</script>
</html>

// --- partB.html (Part 1) ---
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Styled</title>
    <style>
        *{
            margin: 0;
            padding: 0;
            box-sizing: border-box;
            font-family: Arial, sans-serif;
        }
        #nav{
            background-color: blue;
            /* width: 100%; */
            height: 45px;
            color: white;
            /* display: flex; */
        }
        ul{
            list-style: none;
            display: flex;
            height: 100%;
            /* padding-left: 1000px; */
            gap: 15px;
            align-items: center;
            justify-content: flex-end;
        }
        .title{
            width: 100%;
            height: 100px;
            background-color: black;
            color: white;
            text-align: center;
            padding-top: 30px;
        }
        .container{
            display: flex;
            gap: 30px;
            padding: 20px;
            justify-content: center;
        }
        .card{
            width: 500px;
            height: 300px;
            border: 2px solid black;
            border-radius: 15px;
            padding: 30px;
            text-align: center;

```

partB.html - Part 2 (Lines 51-123)

```
        justify-content: center;
        box-shadow: 2px 2px 10px rgba(0, 0, 0, 0.1);
        background-color: #67b46a;

        font-size: 20px;
        display: flex;
        flex-direction: column;
        gap: 15px;
    }
    .btn{
        width: 100px;
        height: 40px;
        border: none;
        border-radius: 5px;
        background-color: blue;
        color: white;
        cursor: pointer;
        transition: all 0.5s ease;
    }
    .btn:hover{
        background-color: rgb(0, 0, 0);
        box-shadow: 0px 0px 20px 10px rgb(229, 255, 0);
        transform: translateY(-3px);
    }
    li{
        cursor: pointer;
        transition: all 0.5s ease;
    }
    li:hover{
        color: rgb(255, 255, 255);
        transform: translateY(-3px);
    }
}
</style>
</head>
<body>
<header>
<div class="title">
<h1>WELCOME TO PAGE</h1>
</div>
<nav id="nav">
<ul>
<li>HOME</li>
<li>CONTEXT</li>
<li>BOOK</li>
<li>NEAR BY</li>
<li>ABOUT US</li>
</ul>
</nav>
</header>
<main class="container">
<article class="card">
<h2>Card One</h2>
<p>This is the first content card. It uses a custom serif font for the heading to stand out from the body text
style.</p>
<center><button class="btn">Read More</button></center>
</article>

<!-- Card 2 -->
<article class="card">
<h2>Card Two</h2>
<p>This is the second content card. Notice the subtle drop shadow and rounded corners applied to the card container
framework.</p>
<center><button class="btn" >Read More</button></center>
</article>

<!-- Card 3 -->
<article class="card">
<h2>Card Three</h2>
<p>This is the third content card. Hover over the button below or the top navigation bar links to see color changes
instantly.</p>
<center><button class="btn">Read More</button></center>
</article>
</main>
</body>
</html>
```

7. CODE EXPLANATION

Code Selector / Statement	Explanation
<code>canvas.getContext("2d")</code>	Retrieves the 2D rendering graphics drawing context bound to the Canvas tag.
<code>ctx.fillRect(20,20,150,200)</code>	Draws a solid filled rectangle shape at x=20, y=20 with width=150 and height=200.
<code>ctx.strokeRect(200,20,150,100)</code>	Draws a rectangle outline using the active stroke styling colors.
<code>ctx.beginPath() / stroke()</code>	Initializes a new vector drawing path / renders the path on screen.
<code>ctx.arc(300,200,50,0,Math.PI*2)</code>	Draws a full 360-degree circle outline at x=300, y=200 with a 50px radius.
<code>display: flex; gap: 30px;</code>	Enables Flexbox layout, aligning cards side-by-side with 30px spacing.
<code>box-shadow: 2px 2px 10px rgba(...)</code>	Applies a soft transparent drop shadow effect underneath room cards.
<code>transition: all 0.5s ease;</code>	Controls the rate of CSS property mutations dynamically on hover.

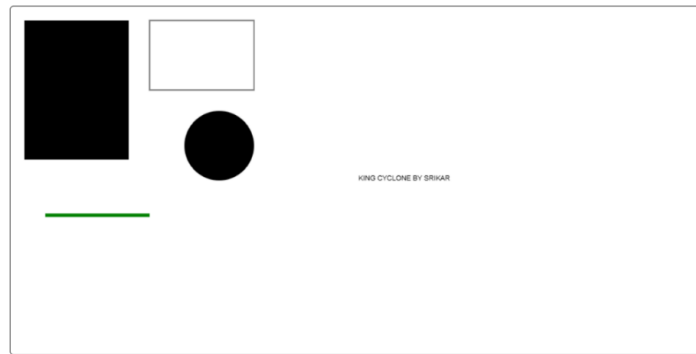
8. TEST CASES AND EXECUTION DATA

The following test executions verify CSS3 layouts and Canvas drawing methods in Google Chrome:

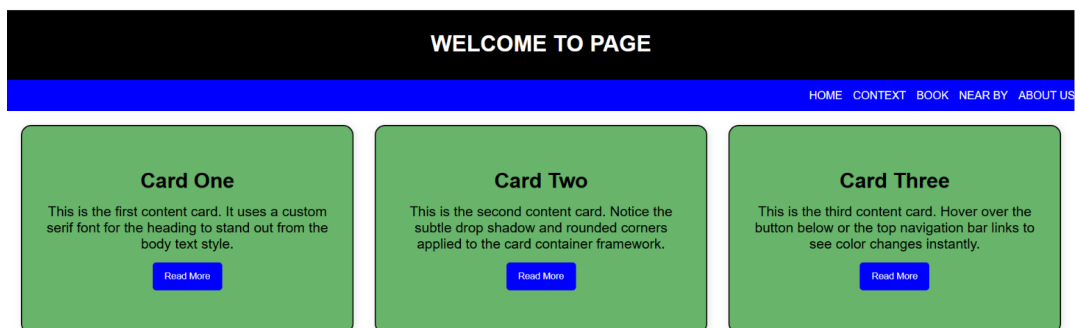
TC ID	Test Description	Input / Action	Expected Result	Status
TC-01	Canvas Shapes Fills	Execute canvas.html	Blue filled rectangle and red rectangle outline render at offsets	Pass
TC-02	Canvas Arc & Paths	Execute canvas.html	Green line and solid black circle render on canvas coordinates	Pass
TC-03	Canvas Text Renders	Execute canvas.html	Text string 'KING CYCLONE BY SRIKAR' displays at x=500, y=250	Pass
TC-04	Flexbox Grid Wrap	Launch partB.html in Chrome	Three cards display aligned horizontally, wrapping on resize	Pass
TC-05	CSS Transitions	Hover mouse over button/links	Button expands, drops shadow, and shifts color smoothly	Pass

9. OUTPUT

Output 1: HTML5 Canvas Elements Rendering (canvas.html)



Output 2: Enhanced CSS3 Card Grid Layout (partB.html)



10. RESULT

Thus, page visual presentation styles using CSS3 (selectors, flex models, borders, box-shadows, animations) and HTML5 inline vector canvas graphics drawings were successfully designed, implemented, and verified.

Submission Package Structure:

Observation/ObTask2/

■ ■ ■ ■ Docs/

■ ■ ■ ■ ObsWeek2.pdf

■ ■ ■ ■ Screenshots/

■ ■ ■ ■ image1.png

■ ■ ■ ■ image2.png

■ ■ ■ ■ html/

■ ■ ■ ■ canvas.html

■ ■ ■ ■ partB.html